

Hardware and forming notes

HARDWARE AND FIT REFERENCES - millimetres

A02 main landing gear: formed strip reference, not a plastic print and not a laser flat blank. Source section is 12.7 mm fore-aft width by 3.175 mm strip thickness. Overall stance approx 285.5 mm; main axle centres $x = \pm 151$ mm with wheels, axle height $z = -135.85$. Use a suitable formed model-aircraft metal gear or commission this geometry. The metalwork shop must select a bendable alloy/temper and finite bend radii appropriate to it, then preserve attachment and axle stations. Source CAD has ideal sharp bends; do not interpret these as zero-radius physical bends. Start with a strip longer than developed centreline, form to the STEP/template and finish ends. Shop calculates bend deductions for its tooling. Two new 3.2 mm attachment holes $x = \pm 22.8222, y = 136.35$ align A47; fit 2xM3 through-bolts, large washers and locking nuts. Suitable small-aircraft landing gear requires physical load/deflection testing for completed aircraft mass.

A46 nose wire: 3 mm spring/piano-wire reference, with upper shaft extended to $z = +1$ mm so it reaches the steering bearing/arm. Preserve the source bends and axle line. Cut to the supplied formed STEP using a bending jig; final upper end can be trimmed only after bearing, shaft collar and steering arm are installed. Wire must rotate freely in finished 3.2 mm bearing bores without lateral play that permits wheel shimmy. Do not 3D print it. Final upper length depends on actual clamp/arm.

Wheels: 3 purchased lightweight 50.8 mm diameter, approximately 16.5 mm wide, 3 mm bore. Two main axle bolts nominal M3x25 mm plus spacers/washers/locking nuts; nose wheel uses wire axle with 3 mm collars. Fit freely rotating wheels without clamping hubs.

Controls: 2x3 mm smooth hinge rods, start 285 mm long, final trim flush to outer end retainers with approximately 1 mm axial freedom. 4 purchased lightweight horns for 2 ailerons + 2 ruddervators; 1 clamping 3 mm nose-steering arm. 5 pushrods with matching clevises/ball links, cut only after neutral servo and surface positions are set. Do not use glue alone to retain hinge pins or wing joiners.

Root wing retention: six M2 through-bolts, start 20 mm length, washers and locking nuts in three aligned root attachment holes per wing. Finish printed/wood holes to the actual fastener clearance. Set lengths to avoid moving linkages. These root fasteners retain removable wings; do not glue the sliding carbon joiner into both wings.

Motor: four M3 bolts through revised A26, 19 mm PCD. Select length to pass 5 mm plywood + washer and engage motor threads without reaching windings. Actual screw engagement depth must be measured; do not blindly reuse a long screw. Manufacturer drawing is supplied.

The 80x35x4 mm print coupon has bores 2.0, 2.2, 3.0, 3.2, 8.0, 8.2, 10.2 mm. Use it to establish printer/material fit; it is not an aircraft part. Size must remain 100%. Verify outside dimensions and fit using intended rods/bolts before printing structural brackets.

Nose bearing A09 to mount A37: four M3 machine screws with washers and locking nuts, start 16-20 mm long and select after measuring the actual stack. The shaft-through portion of the mount is 7 mm and the bearing foot about 3 mm, plus washer/nut stack. Use the printed underside access recesses; do not trap a loose nut where it cannot be inspected. Finish the bearing clearance holes to the actual screw and verify no bolt can touch the rotating nose shaft.